Engineering - Undergraduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. For a short dipole antenna, evaluate the maximum effective apertureA_em.
   (a) 0.119 wavelength
   (b) 0.130 wavelength
   (c) 0.112 $wavelength^2$
   (d) 0.100 wavelength
   (e) 0.123 $wavelength^2$
2. At approximately what temperature will water boil ifthe elevationis 10,000 ft?
   (a) 206
   (a) 193
   (a) 190
   (a) 212
   (a) 180
3. Findcurlaat the point (1, - 2, 1) fora= x^2 y^2 i+ 2 xyzj+ z^2 k(1)
   (a) (4, 4, 0)
   (b) (-2, 0, 0)
   (c) (0, 0, 4)
   (d) (4, 0, 0)
   (e) (2, 0, 0)
4. Using the definition of the unilateral Z transform, F (z) $= ^\infty \sum \_n = 0 f[n]z^-n.findthez-transformofthestep-sequencef[n] = u[n].$

1 / (z - 1)

z / (1 + z)

z^2 / (z - 1)

1 / (2 - z)

(z - 1) / z

A Lissajous pattern on an oscilloscope has 5 horinzontal tangencies and 2 vertical tangencies. The frequency of horizontal input is 100 Hz. The frequency of the vertical will be

(a) 750 Hz.
(b) 250 Hz.
(c) 625 Hz.
(d) 500 Hz.
(e) 800 Hz.

True / False (10 marks)

Answer True or False

6. True or False: The current at time t=0+, i(0^+)$, can be expressed as E/(v(0^+) * R)$.

6. True or False: The variance of the random variable Z = X + Y, regardless of independence, is simply VarZ = VarX + VarY.

7. True or False: The voltage across a linear time-varying inductor can be expressed as U(t) = L_0 (1 - tanh t)cos(t) + L_0 (t - $sech^2$ t)sin(t).

8. True or False: The equivalent resistance of five resistors of 10, 15, 20, 25, and 30 ohms connected in parallel is approximately 3.46 ohms.

9. True or False: Silicon is often regarded as the most harmful impurity in magnetic materials, negatively impacting their performance.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of finding the Laplace transform of the function f(t) = t^n, where n is a positive integer. Explain the significance of the result in engineering applications.

12. Discuss the concept of probability in the context of a single die throw, explaining how the likelihood of rolling a "six" is determined and its implications in engineering applications.

End of Paper